TripSync - A Platform to Synchronize Group Travel

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TripSync - A Platform to Synchronize Group Travel

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1 INTRODUCTION

1.1. PURPOSE

This Software Requirements Specification (SRS) document is aimed at specifying the entire requirements of TripSync A Platform to Synchronize Group Travel. This report describes the functionality of the system, how it will be used, and the limitations within which it will be used. It gives clarity to the developers, stakeholders and to the users in order to make sure the end product meets expectations.
TripSync is a project that will help to streamline the process of planning a group travel by providing a single platform where the user is able to create a trip, book a hotel, handle the expenses, schedule, and communicate in real time. This SRS serves as the foundation for system design, development, and testing.

1.2. PRODUCT SCOPE

TripSync is a comprehensive, all-in-one platform designed to simplify and enhance the experience of planning group trips. It allows users to seamlessly register, create new trips, and join existing trips using unique trip codes, making group coordination easier than ever. Within the platform, users can efficiently manage detailed itineraries, book hotels, track and split expenses, upload receipts, and leave comments or updates for other trip members. By centralizing all these features, TripSync eliminates the need for scattered group chats, cumbersome spreadsheets, and multiple separate booking applications. The system is tailored for families, students, and tour organizers, providing a smooth, intuitive, and collaborative hub for travel planning. With TripSync, organizing group trips becomes not only efficient but also enjoyable, ensuring that every trip runs seamlessly from start to finish.

2 OVERVIEW

THE OVERALL DESCRIPTION

TripSync is an end-to-end solution that allows users to plan and organise group trips. The system enables one to register, create trips, use the trip codes to join the existing trips, manage itineraries, book hotels, divide the expenses, add receipts, and to leave comments to other members.

It aims at providing an easy platform to enhance manual coordination and better travel organization by families, students and tour organizers. It will be able to substitute the old manual

modes, including group chats, spreadsheets, and individual hotel applications, and put it all in one location.

PRODUCT PERSPECTIVE

TripSync is an independent system though it can be connected to third-party services like hotel booking APIs or payment gateways. It is compatible with mobile phones and web browsers, and thus users are able to manage their trips anytime.

The system facilitates the travel process by encouraging teamwork and easy implementation of the tripping process by automation. It fills the gap in the existing travel planning processes by offering a smart and collaborative platform that combines the key management functions.

2.2. PRODUCT FUNCTIONS

Key functions of TripSync include:

- **User Registration and Login:** Allows organizers and travelers to securely create accounts and log in. Ensures data privacy and controlled access to TripSync features.
- **Trip Creation and Joining**: Users can create new trips or join existing ones using unique invitation codes. This enables easy collaboration among trip participants.
- **Itinerary Management:** Users can define travel dates, destinations, and day-by-day plans. All trip members can view updates in real time.
- **Hotel Integration:** Provides options to search, book, and manage hotel reservations within the platform. Users can also edit booking details as needed.
- **Expense Management:** Supports real-time expense tracking with automatic cost splitting. Users can upload receipts for transparency and record keeping.
- **Communication:** Enables group chats and comments for sharing plans and updates. Improves coordination among all trip members.
- **Notifications**: Sends alerts for itinerary changes, reminders, and important updates. Keeps users informed throughout the trip.
- **Role Management:** Displays participant roles such as Organizer or Traveler. Ensures clear responsibilities and controlled access to trip management features.

2.2. USER CHARACTERISTICS

The primary user base for TripSync is diverse, encompassing students arranging group trips, families managing vacation finances, and tour organizers handling complex schedules. Additionally, the platform caters to general travelers seeking a straightforward, low-tech tool for planning. Consequently, the interface is designed for simplicity and intuitive navigation, ensuring that users with varying levels of technical expertise can easily utilize the system without needing advanced skills.

2.3. CONSTRAINTS

The project faces several key constraints. First, a stable internet connection is mandatory to facilitate real-time updates of trip information and data syncing. Second, the system relies heavily on third-party services for essential functions like hotel bookings and payment processing, introducing external dependencies. Third, robust security measures are critical to protect sensitive user data and financial details in compliance with privacy standards. Finally, the application must account for device limitations, as screen size and hardware capabilities on mobile devices can significantly impact layout rendering and overall performance.

2.4. ASSUMPTIONS AND DEPENDENCIES

The system operates on the assumptions that users will provide accurate personal and payment information, that external APIs for hotel booking and payments will remain accessible and stable, and that users will have access to internet-connected devices such as computers or smartphones. Functionally, the project is dependent on reliable connections to booking APIs, payment gateways, and cloud storage services. It also requires a scalable database to manage trip, user, and expense records, along with a notification service to deliver timely updates and reminders via push or email.

3 STATE OF THE ART

LITERATURE REVIEW

Several existing travel platforms attempt to simplify trip planning, yet most focus on individual rather than group needs. For example, TripIt helps users organize personal itineraries by compiling flights and hotel bookings but lacks collaborative features such as shared editing or expense management. Google Travel provides destination insights and automated suggestions, but it does not support real-time group coordination or communication within a shared trip. Airbnb offers accommodation booking and allows users to share trip details, yet it still does not include tools for collective planning, budgeting, or daily schedule management. Many travelers also rely on separate apps like Splitwise for expenses and WhatsApp for communication, which leads to scattered information and difficulty keeping everyone updated.

These limitations highlight a major gap: no single platform provides integrated tools for collaboration, messaging, itinerary updates, and expense tracking. TripSync aims to address this gap by offering a unified system where groups can plan trips together, manage expenses, search hotels, update itineraries, and communicate in real time, making it more effective for coordinated group travel than the existing solutions.

EXISTING SYSTEMS

Several travel-planning systems exist, such as:

1. **TripIt**
   Organizes travel bookings into a single, easy-to-view itinerary.
2. **Airbnb Trips**
   Enables booking accommodations and local activities in one platform.
3. **Google Travel**
   Help users plan trips, track bookings, and get suggestions.
4. **Group travel apps**
   Provide basic shared itineraries and communication with limited features.

These systems offer useful features like booking, itinerary viewing, and reminders.

System	Strengths	Limitations
TripIt	• Clean user interfaces • Integration with hotel and flight data	• Limited group collaboration features • Often lacks expense-splitting tools
Airbnb Trips	• Destination suggestions and itinerary generation • Personal trip organization	• No centralized platform combining communication + booking + expense management + itinerary editing
Google Travel	• Integration with hotel and flight data • Destination suggestions	• Weak or no real-time syncing when plans change • Not designed for university trips or organized groups needing shared access
Group Travel Apps	• Personal trip organization • Clean user interfaces	• Limited functionality beyond basic shared itineraries • Often lack robust expense management

4 USER/SYSTEM REQUIREMENTS

External Interface Requirements

4.1.1 User Interfaces

The User Interface (UI) is developed using **React** and styled with **Tailwind CSS** to provide a highly responsive, modern, and component-based experience. The interface is designed to support real-time collaboration without requiring manual page refreshes.

Key Screens and Features:

- **Authentication Portal:** A secure login modal utilizing Firebase UI for Google OAuth and Email/Password sign-in.
- **Trip Dashboard:** A central hub displaying active trips, utilizing real-time listeners to update data instantly when other users make changes.
- **Interactive Itinerary:** A drag-and-drop interface (implemented via react-beautiful-dnd or similar) allowing users to visually reorganize activities across days.
- **Live Hotel Search:** A search interface directly integrated with the **Booking.com API**, displaying hotel cards with images, ratings, and "Add to Itinerary" buttons.
- **Expense Tracker:** A detailed list view where users can log expenses, upload receipts (handling image previews), and view a dynamically calculated "Split Summary."
- **Real-Time Chat:** A sliding side-panel or embedded chat window displaying message bubbles, user avatars, and timestamps, synced via Cloud Firestore.

UI Standards:

- **Responsive Design:** Utilizes Tailwind's utility classes (e.g., md:flex, lg:w-1/2) to ensure usability across mobile phones, tablets, and desktops.
- **Feedback Mechanisms:** Toast notifications (e.g., "Trip Saved", "Link Copied") provide immediate system feedback for user actions.

4.1.2 Hardware Interfaces

TripSync is a web-based application and does not require proprietary hardware. It interacts with standard device capabilities:

- **Client Device:** Any device (Smartphone, Tablet, Laptop, Desktop) capable of running a modern web browser (Chrome, Firefox, Safari, Edge).
- **Network Adapter:** Requires an active internet connection (Wi-Fi or Cellular Data) to maintain the persistent WebSocket connection with Firebase.
- **Peripherals:** Support for touchscreens (for drag-and-drop on mobile) and cameras/file-systems (for uploading receipt images).

4.1.3 Software Interfaces

The system relies on the following specific software components and APIs:

- **Frontend Framework: React (Vite)** is used for rendering the View layer, managing component state, and handling client-side routing.
- **Backend / Database: Google Cloud Firestore** serves as the NoSQL database, storing user data, trip metadata, and chat logs in JSON-like documents.
- **Authentication Service: Firebase Authentication** handles identity management, token generation, and session security.
- **File Storage: Firebase Storage** is used to host user-uploaded content, specifically images of payment receipts.

- **External Data API: Booking.com API (via RapidAPI)** is consumed to fetch real-time hotel availability, pricing, and location data.

4.1.4 Communication Interfaces

- **Communication Protocols:**
  - **HTTPS (TLS 1.2+):** Used for all initial page loads and REST API calls (e.g., fetching hotel data from RapidAPI) to ensure data confidentiality.
  - **WebSockets (WSS):** Utilized implicitly by the Firestore SDK to maintain a live, bi-directional connection for pushing updates (chat messages, itinerary changes) to connected clients instantly.
- **Data Format:** JSON (JavaScript Object Notation) is used for all data exchange between the client, Firebase, and external APIs.

5 Functional Requirements

The following section outlines the specific functional requirements for TripSync.

Functional Requirements with Traceability information

Each requirement will have a separate table

Requirement ID	FR-01		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-01
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***Agreed to***
Parent Requirement #	N/A								
Description	The system must allow secure login and logout for all users.								
Rationale	To ensure only authorized users can access the system's features.								
Source	SRS Part 1 – Section 2.				**Source Document**		-		
Acceptance/Fit Criteria	Users can login using valid credentials and are denied access with invalid ones.								
Dependencies	Database connectivity, user authentication mechanism								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***		
Change History	New								

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Requirement ID	FR-02		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-02
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***Agreed to***
Parent Requirement #	N/A								
Description	The system must allow group coordinators to create a travel plan and obtain a join code to invite members.								
Rationale	Core app functionality								
Source	Scope Document				**Source Document**		-		
Acceptance/Fit Criteria	Users create trips with required fields and system generates a unique join code.								
Dependencies	Trip database, code generation algorithm								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***		
Change History	New								

Requirement ID	FR-03		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-03
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***New***
Parent Requirement #	FR-03								
Description	The system must allow other group members to join an existing trip using a join code								
Rationale	Enables group membership and collaboration								
Source	Scope Document				**Source Document**		-		

Acceptance/Fit Criteria	Valid code adds user to trip roster, and invalid code shows error message.						
Dependencies	FR-002, invite code validation algorithm						
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***
Change History	New						

Requirement ID	FR-04		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-04
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***new***
Parent Requirement #	N/A								
Description	The system shall provide a chat thread and comments for each trip.								
Rationale	Enables coordination and quick group communication								
Source	Scope Document				**Source Document**		-		
Acceptance/Fit Criteria	Messages displayed with sender and time with real time communication.								
Dependencies	Messaging service, file storage								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***		
Change History	New								

Requirement ID	FR-05		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-05
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***New***
Parent Requirement #	N/A								
Description	The system shall track trip expenses, add/edit items, mark items as paid/unpaid, split expenses across members and show total expenses								
Rationale	All travel members can see their individual share of expense without need to manually check.								
Source	Scope Document				**Source Document**		-		

Acceptance/Fit Criteria	Add expense with name and amount and the system computes share per member. The UI shows each user's balance.						
Dependencies	Trip database, file storage and currency formatter.						
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***
Change History	New						

Requirement ID	FR-06		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-06
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***new***
Parent Requirement #	N/A								
Description	The system shall accept card payments through verifying card details.								
Rationale	Allows user to pay trip expenses, bookings, and deposits.								
Source	Scope Document				**Source Document**		-		
Acceptance/Fit Criteria	Payments verified and successful payment status is shown.								
Dependencies	Backend payment API								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***		
Change History	New								

Requirement ID	FR-07		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-07
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***new***
Parent Requirement #	N/A								
Description	The system shall search hotels for trips, show prices and allow booking for the hotels.								
Rationale	Convenience for travelers to book hotel while managing trip schdule								
Source	Scope Document				**Source Document**		-		

Acceptance/Fit Criteria	Hotel listed with price and amenities. Booking through entering details and booking confirmation saved for trip						
Dependencies	External hotel API						
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***
Change History	New						

Requirement ID	FR-08		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-08
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***new***
Parent Requirement #	N/A								
Description	The system shall display day by day flight departure, hotel check-in and activities notification and allow coordinator to edit these.								
Rationale	Clear schedule for all members improves planning								
Source	Scope Document				**Source Document**		-		
Acceptance/Fit Criteria	Entries are displayed in order. Coordinator can add/edit and changes sync to members in real time.								
Dependencies	Trip database, notifications								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***		
Change History	New								

Requirement ID	FR-09		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-09
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***new***
Parent Requirement #	N/A								
Description	The system shall allow coordinators to set reminders and send in-app notifications.								
Rationale	Keep members informed of every event timely.								
Source	Scope Document				**Source Document**		-		

Acceptance/Fit Criteria	Reminders can be created with date/time recipients, and a notification pop up is displayed to the other group members.						
Dependencies	Push notification service						
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Optional***
Change History	New						

Requirement ID	FR-10		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-10
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***New***
Parent Requirement #	N/A								
Description	The system shall allow users to edit profiles and update their payment info.								
Rationale	Users need to keep contact/payment details updated.								
Source	Scope Document				**Source Document**		-		
Acceptance/Fit Criteria	Profile fields are editable and this update is reflected in trips.								
Dependencies	Auth, payment gateway tokens								
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Optional***		
Change History	New								

Requirement ID	FR-11		**Requirement Type**		FUNCTIONAL		**Use Case #**		UC-11
Status	***New***		***Agreed-to***	-	***Baselined***	-	***Rejected***	-	***New***
Parent Requirement #	N/A								
Description	The system must provide explicit logout button which ends session.								

Rationale	Security and common user function						
Source	Scope Document		**Source Document**		-		
Acceptance/Fit Criteria	Logout invalidates token server-side and user is redirected to login page.						
Dependencies	Auth backend						
Priority	***Essential***		***Conditional***	-	***Optional***	-	***Essential***
Change History	New						

6 Non-Functional Requirements & System Attributes

6.1 Performance Requirements

- Api latency for core endpoints should be less than 200ms under normal load.
- Messages should appear to recipients within 1 second under normal condition for users in same region.
- Payment gateway integration must support at least 20 payment processing operations per second.
- Search for hotels and listing should respond within 1.5 seconds
- Mobile app start time should be less than 2.5 seconds

6.2 Performance Reliability

- 99.9% uptime for the application services
- Daily backups should be kept for all trip, expenses and booking data

6.3 Security

- OAuth with short lived tokens
- Using PCI compliant payment gateway and store only card tokens

- TLS for all network traffic
- Follow OWASP guidelines for mobile app security
- Data passwords encrypted

7 Project Design/Architecture

1. 4+1 ARCHITECTURE VIEW MODEL (mandatory for Software Projects)

7.1 Case View

Figure 1: Use-case Diagram

Use case Description

Use case ID	UC-1
Use Case Name	Sign Up

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Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Sign Up button
Description	This use case allows a new user to create an account by providing required personal details to access TripSync services.
Preconditions	User is not registered
Postconditions	Account created successfully
Normal Flow	1. Open app 2. Click Sign Up 3. Enter details 4. Submit
Alternate Flow	If details are invalid or missing, system shows error message

Use case ID	UC-2
Use Case Name	Sign In
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	Click Sign In button
Description	Allow a registered user to log in securely using valid credentials.
Preconditions	User is already registered Valid login credentials are available
Postconditions	User is authenticated User is redirected to dashboard
Normal Flow	1. Enter credentials 2. Click Sign In 3. Enter details 4. Submit
Alternate Flow	If invalid details, system prompts user

Use case ID	UC-3
Use Case Name	Create Trip
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the create Trip button
Description	Allow a user to create a new trip by providing destination and travel details.
Preconditions	New trip is created Trip becomes available to participants
Postconditions	Account created successfully
Normal Flow	User clicks Create Trip User enters trip details System saves the trip
Alternate Flow	If required fields are missing, system prompts user

Use case ID	UC-4
Use Case Name	Join Trip
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User enters the trip code
Description	Allows a user to join an existing trip using a valid trip code.
Preconditions	User is logged in Valid Trip code is available
Postconditions	User is added to the trip

Normal Flow	User enters trip code System validates and adds user
Alternate Flow	If code is invalid, system shows error

Use case ID	UC-5
Use Case Name	View Trip Details
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Open trip button
Description	Allows the user to view details of a selected trip
Preconditions	User is a member of the trip
Postconditions	Trip information is displayed
Normal Flow	• User selects a trip • System displays trip details
Alternate Flow	If user is not authorized, access is denied

Use case ID	UC-6
Use Case Name	View Itinerary
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Itinerary button
Description	Allows the user to view the travel itinerary and scheduled activities
Preconditions	Trip itinerary exists

Postconditions	Itinerary is displayed
Normal Flow	• User opens itinerary section • System shows itinerary
Alternate Flow	If itinerary is empty, system shows notification

Use case ID	UC-7
Use Case Name	Search Hotels
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Search button
Description	Allows the user to search hotels based on location and dates
Preconditions	Internet connection is available
Postconditions	List of available hotels is displayed
Normal Flow	• User enters search criteria • System displays hotel options
Alternate Flow	If no hotels are found, system shows message

Use case ID	UC-8
Use Case Name	Book Hotel
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Book button
Description	Allows the user to book a selected hotel for the trip

Preconditions	• Hotel is selected • Payment method is added
Postconditions	Hotel booking is confirmed
Normal Flow	• User confirms booking • System processes the booking
Alternate Flow	If payment fails, booking is cancelled

Use case ID	UC-9
Use Case Name	Add Expense
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Add Expense button
Description	Allows the user to add shared expenses for the trip
Preconditions	• Trip exists • User is a trip member
Postconditions	Expense is recorded in the system
Normal Flow	• User enters expense details • System saves the expense
Alternate Flow	If amount is invalid, system shows error

Use case ID	UC-10
Use Case Name	Split Expenses
Created By	Bushra Naz
Date Created	07 December 2025

Actor	User
Trigger	User clicks the Split Expense button
Description	Allows the user to split expenses among trip members
Preconditions	• Expense exists • Multiple members are present
Postconditions	Expense is divided among selected members
Normal Flow	• User selects members • System calculates split amounts
Alternate Flow	If members are not selected, system prompts user

Use case ID	UC-11
Use Case Name	Add Payment Method
Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	User clicks the Add Payment button
Description	Allows the user to add a payment method for transactions
Preconditions	User is logged in
Postconditions	Payment method is saved securely
Normal Flow	• User enters payment details • System validates and saves method
Alternate Flow	If details are invalid, system shows error

Use case ID	UC-12
Use Case Name	View Dashboard

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Created By	Bushra Naz
Date Created	07 December 2025
Actor	User
Trigger	Users open the dashboard
Description	Allows the user to view an overview of trips, expenses, and activities
Preconditions	User is logged in
Postconditions	Dashboard information is displayed
Normal Flow	• User opens dashboard • System loads summary data
Alternate Flow	If data fails to load, system shows error

Use case ID	UC-13
Use Case Name	Manage Itinerary
Created By	Bushra Naz
Date Created	07 December 2025
Actor	Admin
Trigger	Admin clicks the Manage Itinerary button
Description	Allows the admin to manage and update trip itinerary details
Preconditions	• Admin privileges are granted • Trip exists
Postconditions	Itinerary is updated successfully
Normal Flow	• Admin edits itinerary • System saves changes
Alternate Flow	If invalid data is entered, system rejects update

Use case ID	UC-14
Use Case Name	Add Trip
Created By	Bushra Naz
Date Created	07 December 2025
Actor	Admin
Trigger	Admin clicks the Add Trip button from the admin dashboard
Description	Allows the admin to create and add a new trip by entering complete trip details so that users can join and plan travel activities.
Preconditions	• Admin is logged into the system • Admin has permission to manage trips
Postconditions	• A new trip is successfully created • Trip is saved in the system and visible to users
Normal Flow	• Admin selects the Add Trip option • Admin enters destination, dates, and other details • System validates and saves the trip
Alternate Flow	If required details are missing or invalid, the system prompts the admin to correct them

Use case ID	UC-15
Use Case Name	Edit Trip
Created By	Bushra Naz
Date Created	07 December 2025
Actor	Admin
Trigger	Admin clicks the Edit button
Description	Allows the admin to modify existing trip information to ensure details remain accurate and up to date.
Preconditions	• Admin is logged in • The selected trip already exists
Postconditions	• Trip information is updated successfully • Updated details are reflected for all trip members

Normal Flow	• Admin selects a trip to edit • Admin modifies required fields • System saves the updated trip details
Alternate Flow	If invalid data is entered, the system displays an error and rejects the update

Use case ID	UC-16
Use Case Name	Delete Trip
Created By	Bushra Naz
Date Created	07 December 2025
Actor	Admin
Trigger	Admin clicks the Delete button
Description	Allows the admin to permanently remove a trip from the system when it is no longer needed
Preconditions	• Admin is logged into the system • The selected trip exists
Postconditions	• Trip is removed from the system database • Trip data is no longer accessible to users
Normal Flow	• Admin selects the trip to delete • Admin confirms the deletion action • System deletes the trip
Alternate Flow	If admin cancels the deletion, the system retains the trip without changes

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Logical View

Class Diagram

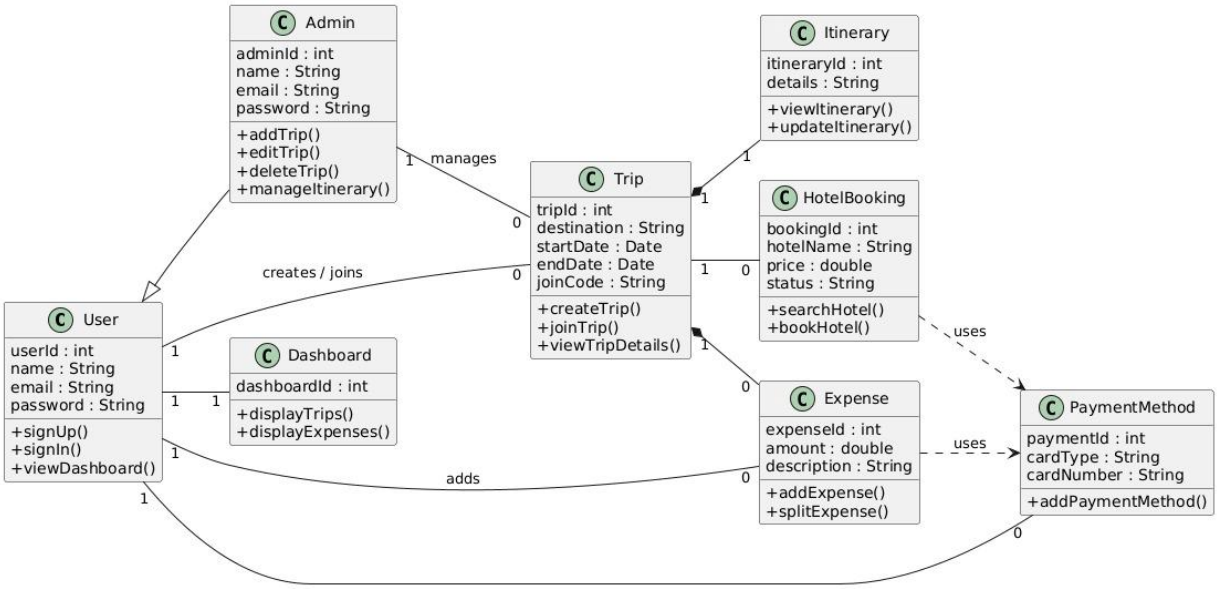


Figure 2: Class Diagram

Sequence Diagram

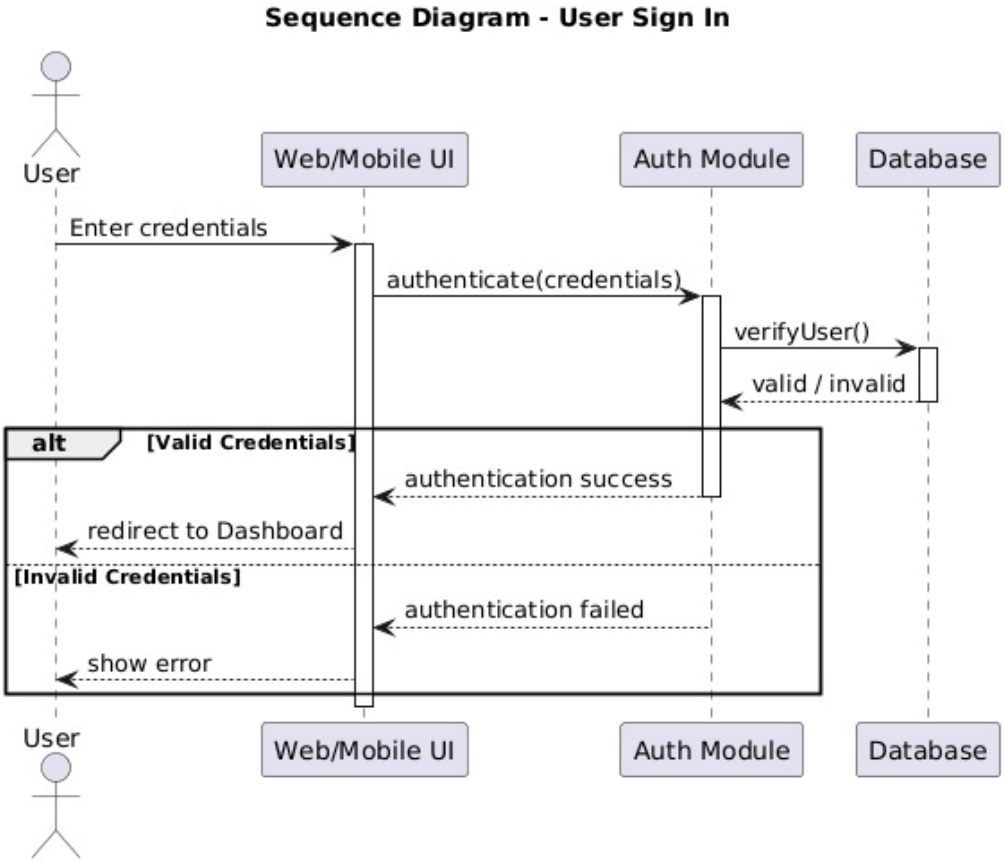


Figure 3: User Sign in

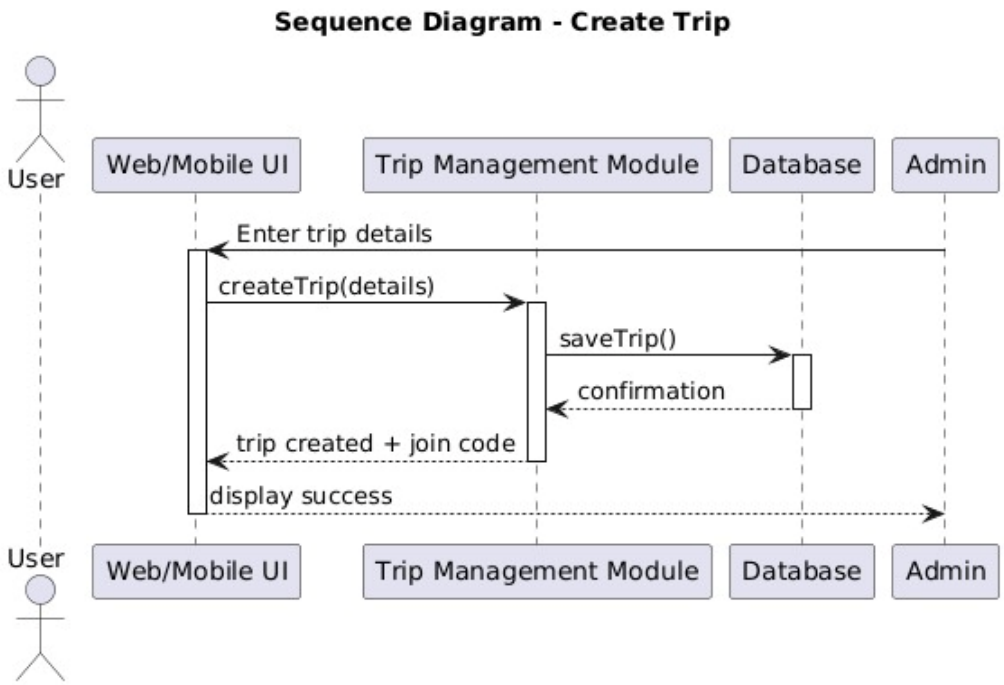


Figure 4: Create Trip

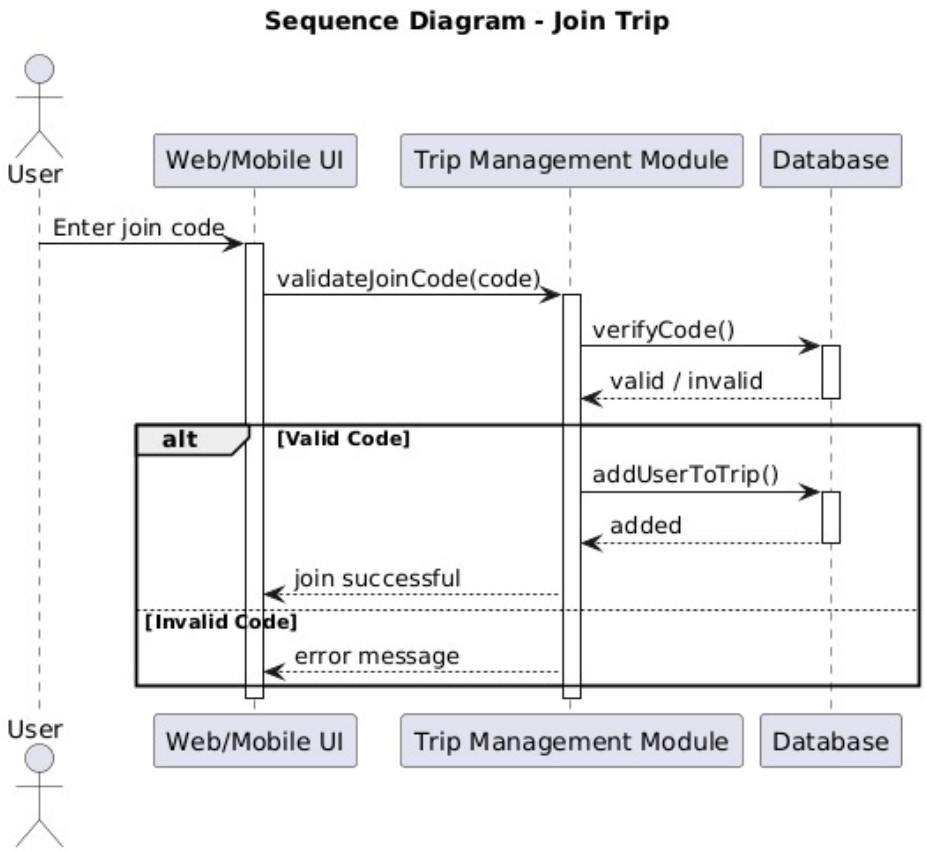


Figure 5: Join Trip

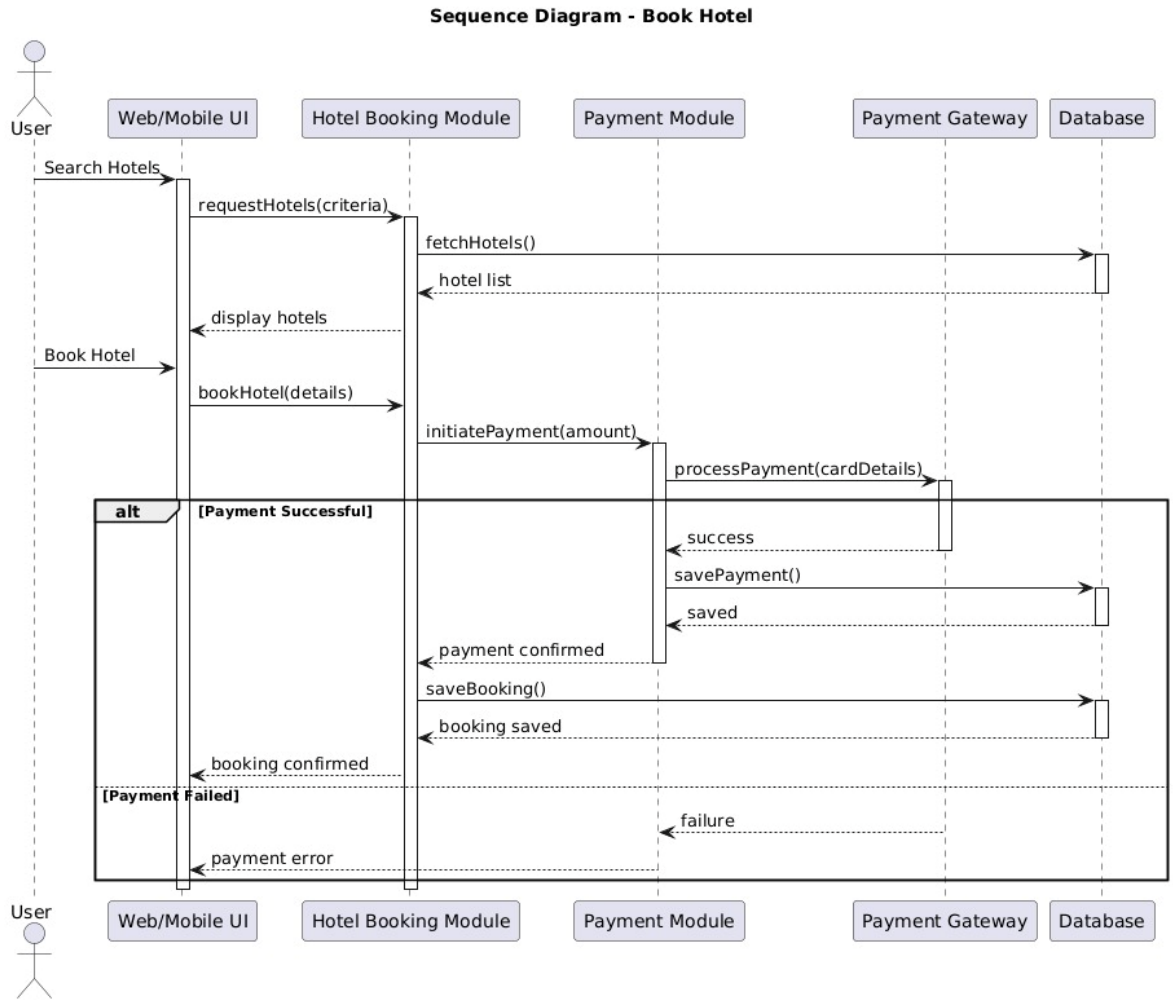


Figure 6: Hotel Booking

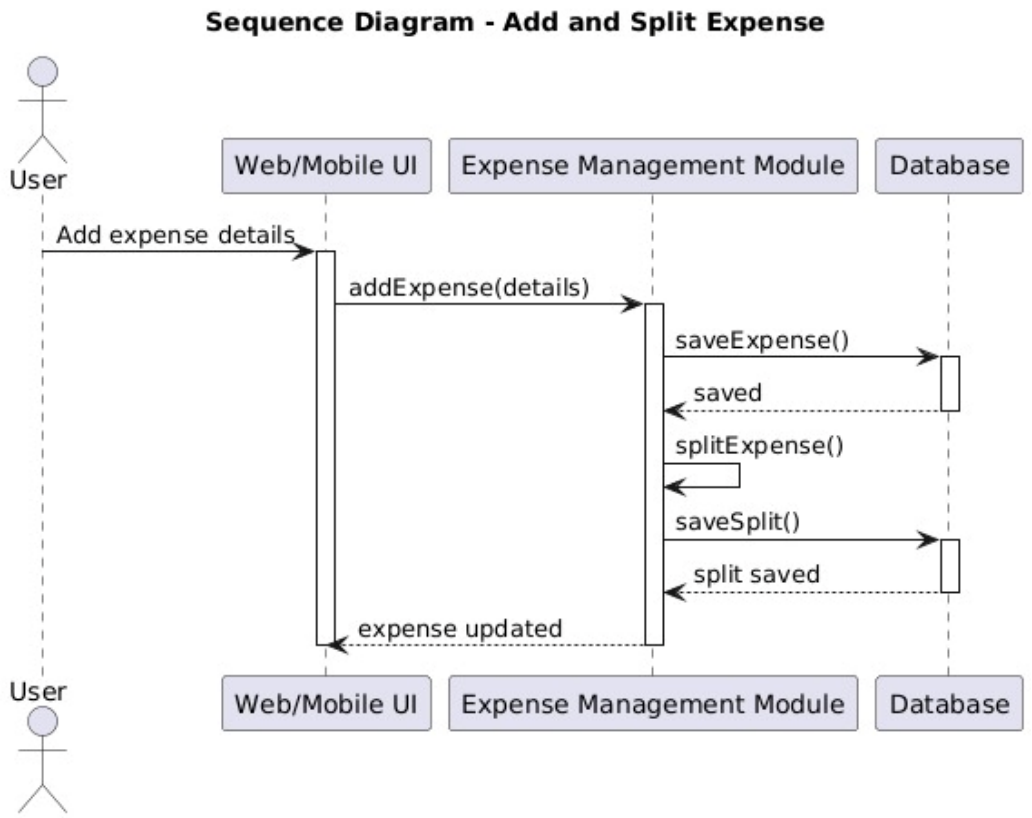


Figure 7: Add and Split Expense

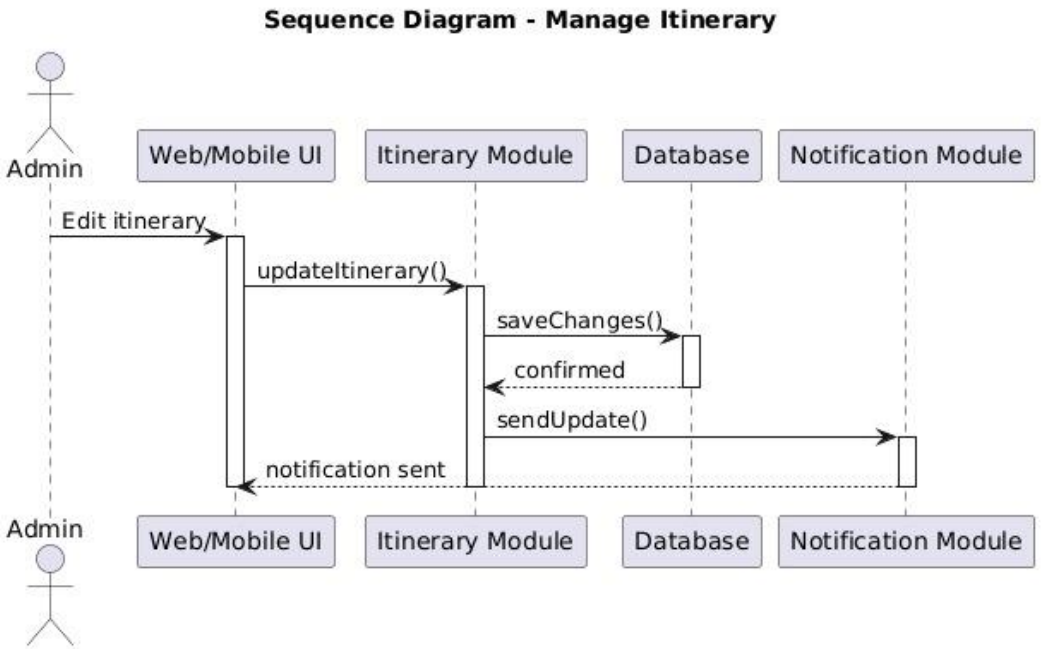


Figure 8: Manage Trip

7.3 Process View

Activity Diagrams

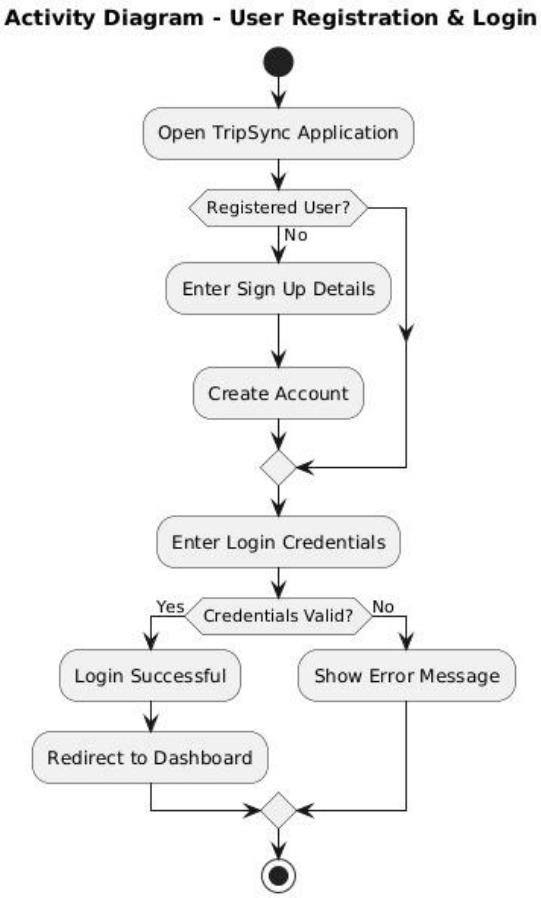


Figure 9:User Registration and Login

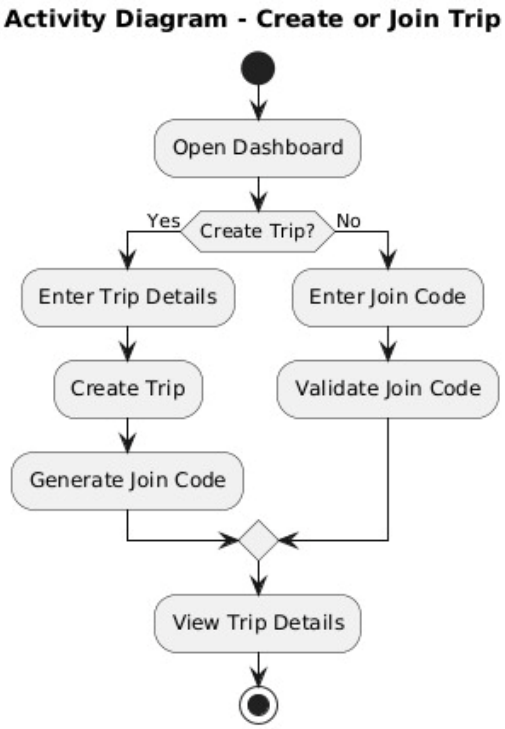


Figure 10: Create or Join Trip

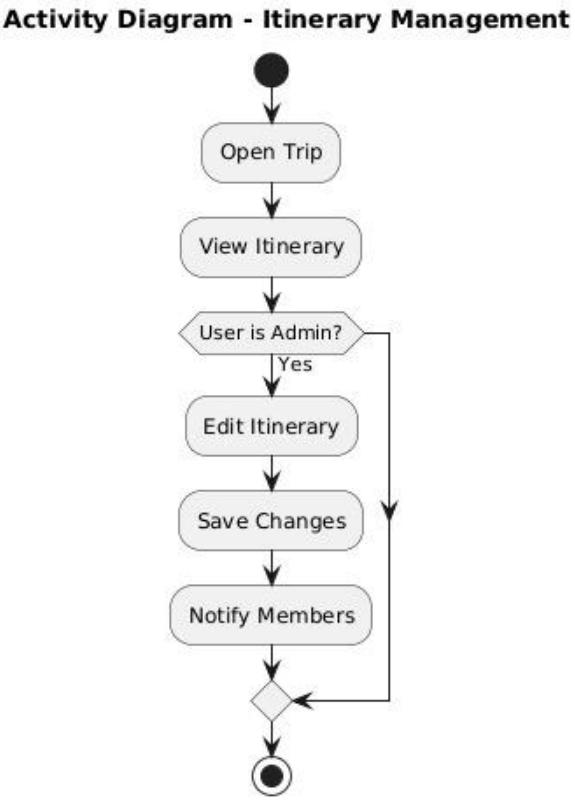


Figure 11: Trip Management

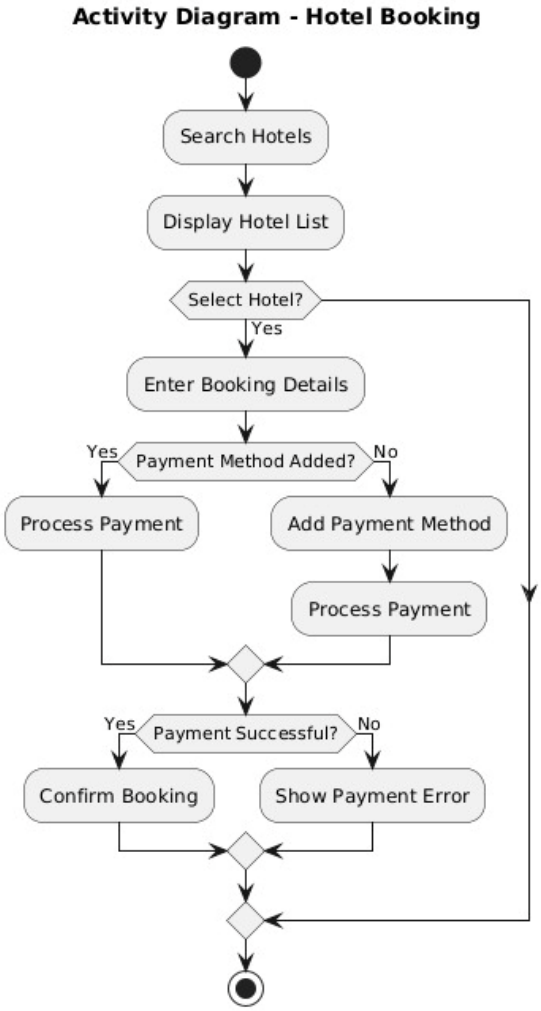


Figure 12: Hotel Booking

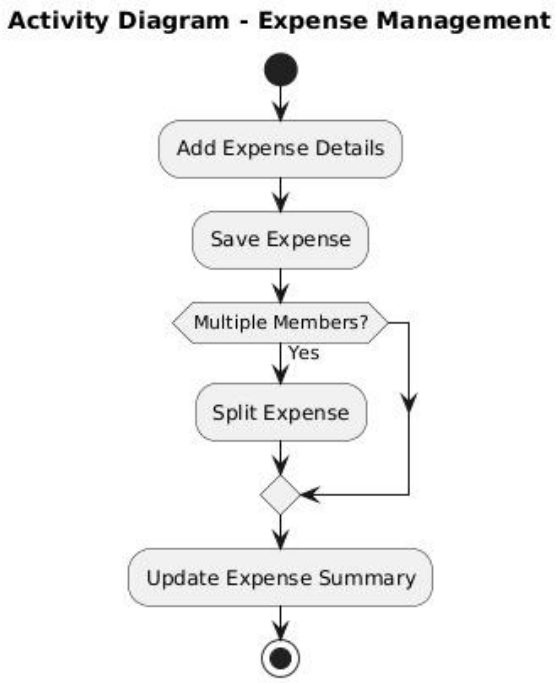
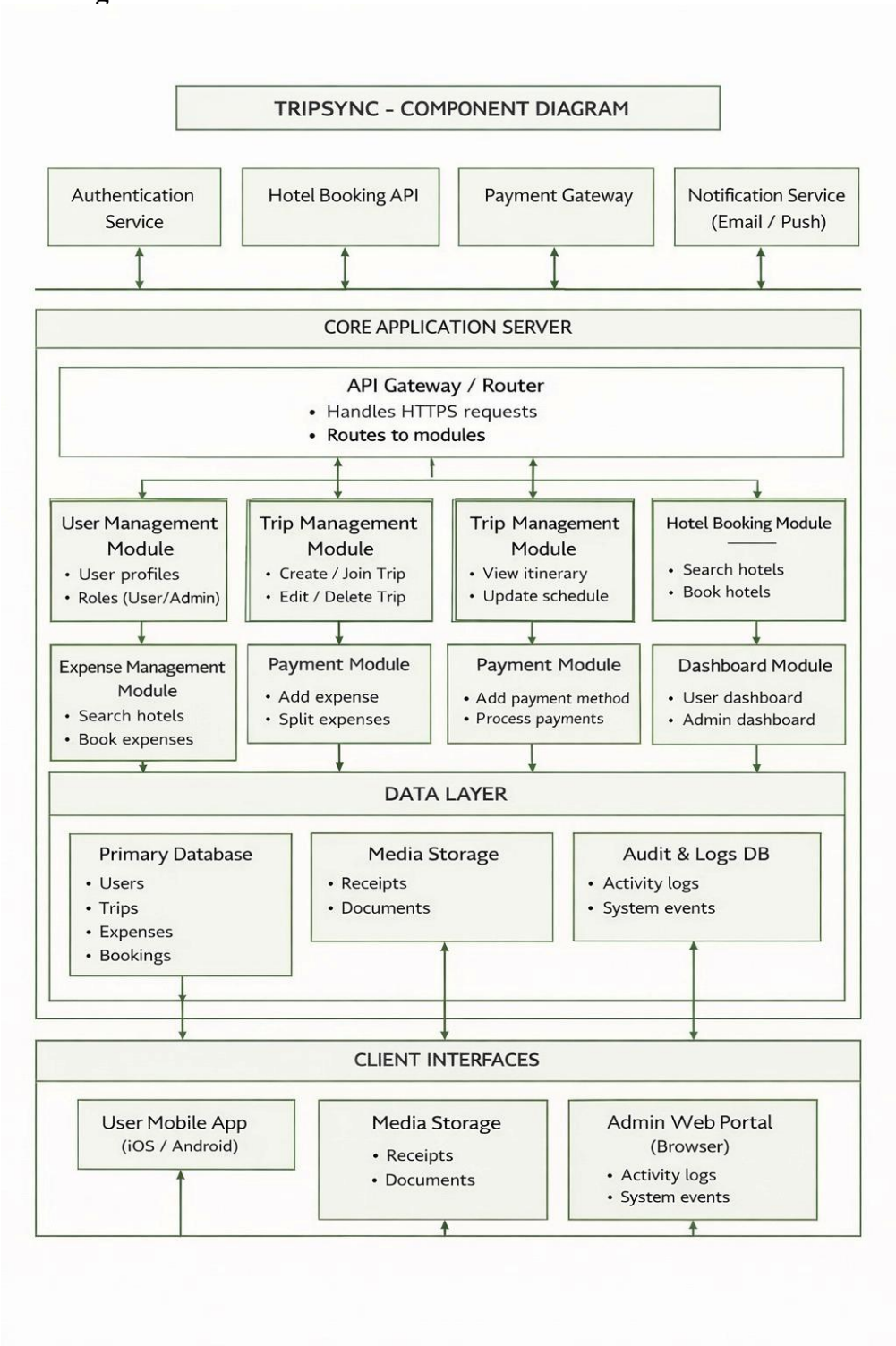


Figure 13: Expense Management

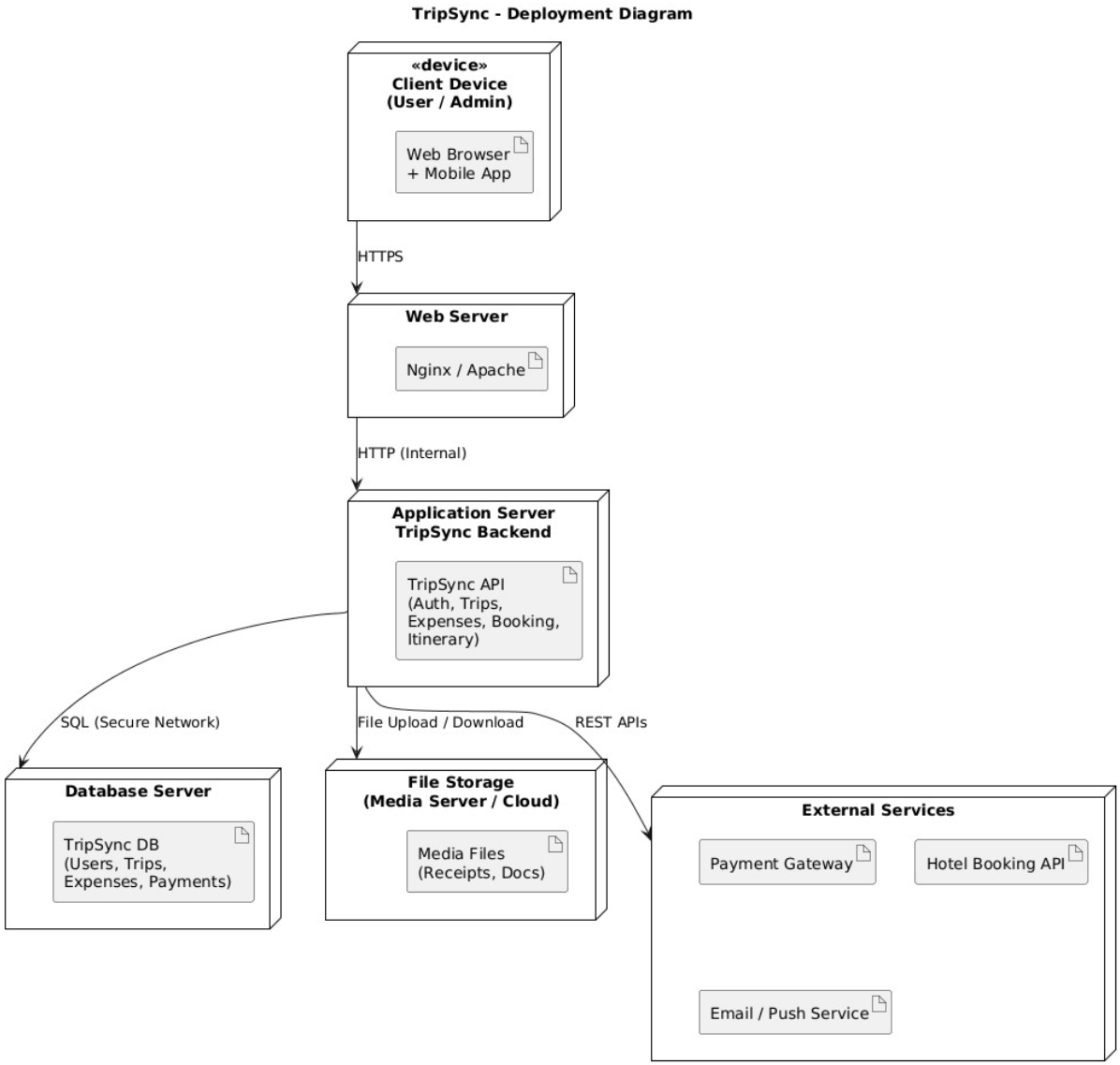
7.4 Development View

Component Diagram:



7.5 Physical view

Deployment Diagram:



User Interface Design

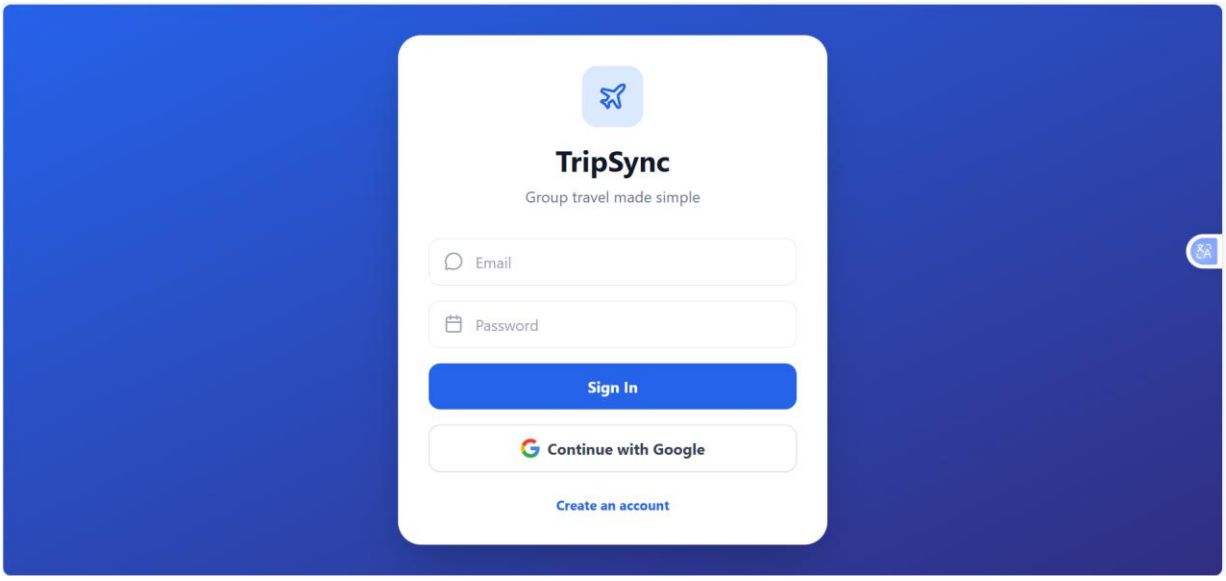


Figure 1: Registration and Login Screen

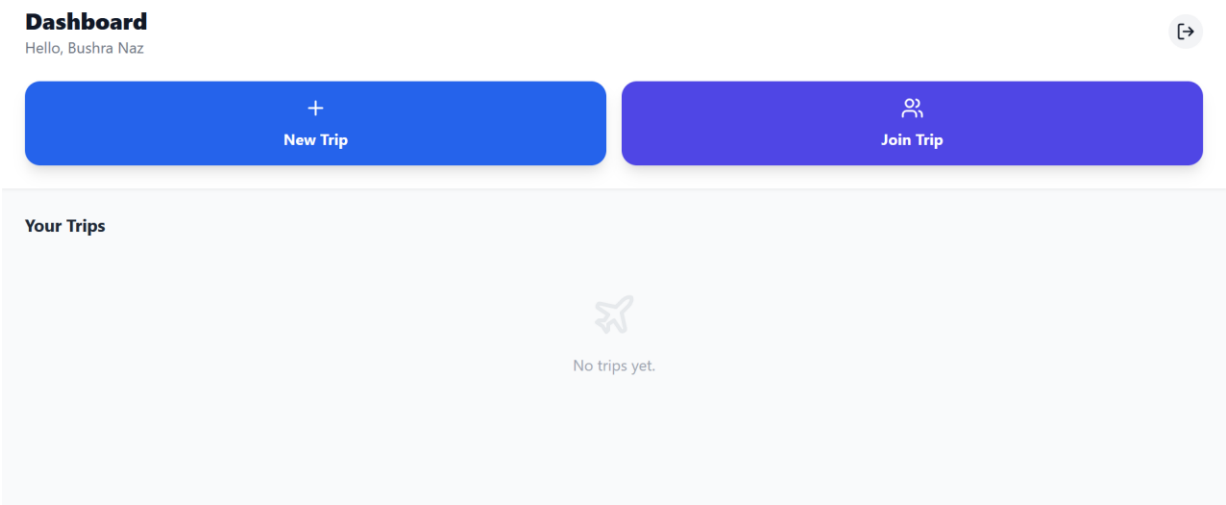
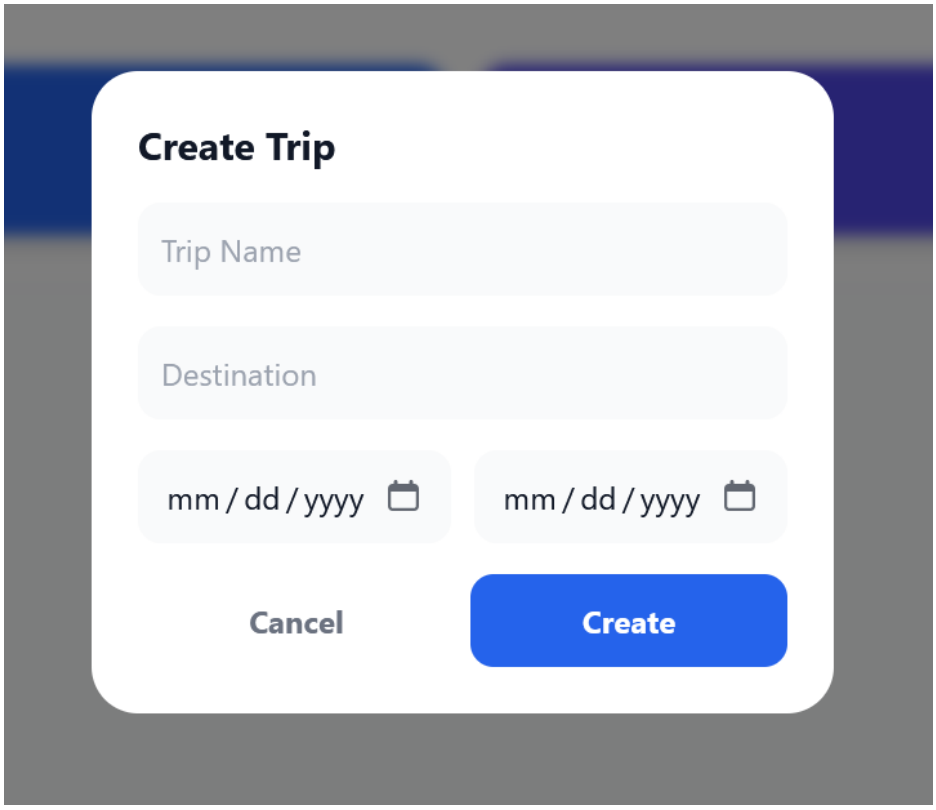
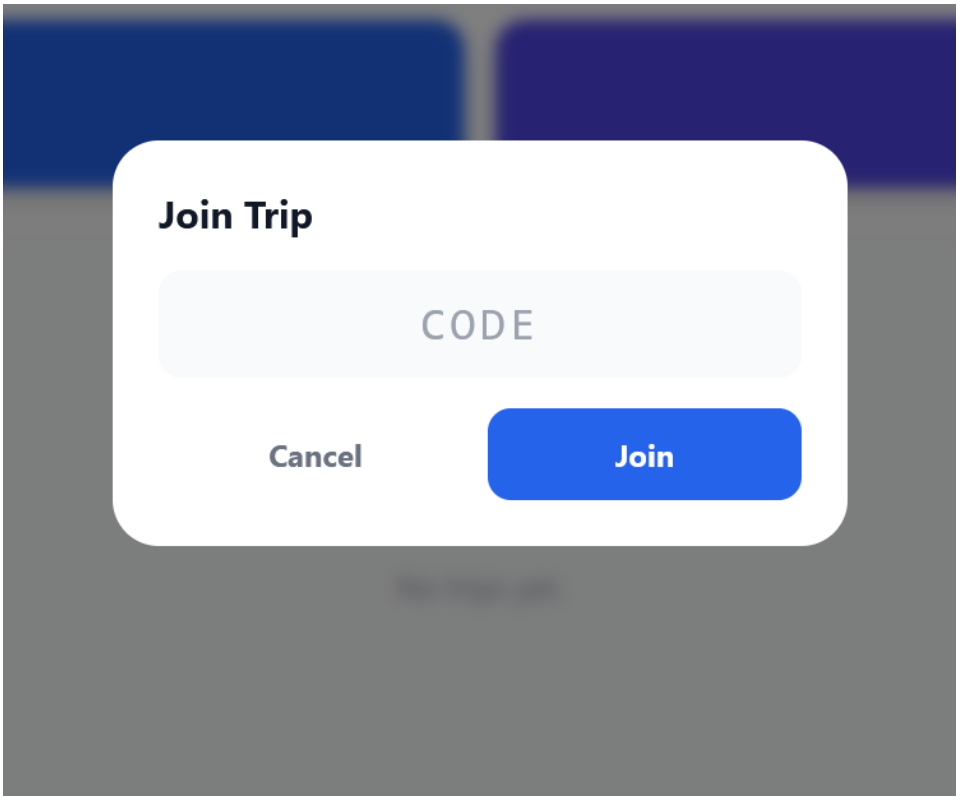


Figure 2: Create and Join Trip Dashboard


Create Trip
Trip Name
Destination
mm / dd / yyyy
mm / dd / yyyy
Cancel
Create


Join Trip
CODE
Cancel
Join

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